

MS&E 228 (Winter 2026) — Lecture 8 Notes (Student Handout)

Deep Dive into OLS Coefficients and the Double Machine Learning Estimator

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Readings. *Applied Causal Inference Powered by ML and AI*, §1, 4, 9. Relevant papers: Frisch & Waugh (1933), Lovell (1963), Chernozhukov et al. (2018).

1 Introduction: Recapping Continuous Treatments and OLS

In our previous discussion, we began exploring the challenges of causal inference with continuous treatments, a scenario frequently encountered in real-world data analysis. We established that continuous treatments are inherently more complex than their binary counterparts. Unlike binary treatments, we cannot invoke a simple analogue of the doubly robust estimator without making additional assumptions or exploring alternative analytical paths. Two potential approaches were briefly mentioned: the binning method (where we discretize the treatment and analyzing the setting similar to the binary case) and making certain modeling choices.

We then proceeded down the path of modeling the relationship between the outcome and the continuous treatment. We started with the most straightforward assumption: a linear relationship. Specifically, we assumed that the outcome is a linear function of both the treatment and a set of control variables. Under this assumption, we found that we could identify the marginal effect of the treatment by examining the coefficient associated with the treatment variable in an Ordinary Least Squares (OLS) regression. However, this identification relies on the crucial assumption that the coefficient associated with the treatment variable in this OLS specification is unique.

This leads to a more fundamental set of questions about OLS itself. What does a coefficient in a multivariate regression actually represent? The common interpretation is that for a one-unit increase in a variable, the outcome changes by the coefficient amount, holding all other variables constant. While this interpretation is useful, it raises a deeper question: what happens if the underlying model is not truly linear? This highlights the need for a more robust understanding of OLS coefficients. By the end of this chapter, you will be equipped to provide a precise and insightful interpretation of OLS coefficients, understand the conditions for their uniqueness, and

analyze the sources of their variance. This chapter is structured around four key goals:

1. **Develop deeper intuition for OLS coefficients:** What does a coefficient in a multivariate regression really mean? When is it uniquely identified, and what drives its variance?
2. **Discover another “insensitive” formula:** Can we simplify inference for a target coefficient in an OLS specification by reducing it to a “uni-variate analysis” ignoring estimation error of other coefficients in the OLS model?
3. **Move beyond full linearity:** How can we allow flexible ML models for covariates while still getting valid causal inference on treatment effects?
4. **Build a practical algorithm for continuous treatments:** What estimator can we use in practice, and what does it estimate when treatment effects are heterogeneous?

By the end of this chapter, we will have answered all four questions through two marvelous theorems—the Frisch-Waugh-Lovell theorem and the Insensitivity theorem—and developed the Double Machine Learning algorithm that is now widely used in industry.

2 Understanding OLS Coefficients via the Frisch-Waugh-Lovell Theorem

We begin by revisiting the four questions posed at the end of the previous chapter regarding the OLS coefficient θ associated with D in the regression $Y \sim D, X$:

1. How can we interpret θ intuitively?
2. When does θ have a unique solution?
3. What drives high variance in the estimate $\hat{\theta}$?
4. What is the analogue of “overlap” under linearity?

This chapter will answer all four of these questions, starting with a foundational result in econometrics: the Frisch-Waugh-Lovell (FWL) theorem. We will answer these questions in the absence of any assumptions on how the world actually works, i.e., we will be making no assumptions on the conditional expectation function $\mathbb{E}[Y \mid D, X]$; we will not assume it to be linear, even though we are fitting a linear model.

2.1 The Frisch-Waugh-Lovell (FWL) Theorem

The FWL theorem provides a powerful and intuitive interpretation of the coefficients in a multivariate regression. It states that a complex, high-dimensional regression can be broken down into a series of simpler, more interpretable steps.

Theorem (Frisch-Waugh-Lovell)

The coefficient $\hat{\theta}$ on the treatment variable D in the multivariate OLS regression of Y on D and X , denoted $\text{OLS}(Y \sim D, X)$, is numerically identical to the coefficient obtained from the following three-step procedure:

1. **Residualize the outcome:** Regress the outcome Y on the covariates X and compute the residuals. These residuals, which we denote $\check{Y}$, represent the part of the outcome that is not linearly predictable from the covariates—the “surprise” in Y .

$$\check{Y} = Y - \text{OLS}(Y \sim X).\text{predict}(X)$$

2. **Residualize the treatment:** Regress the treatment D on the covariates X and compute the residuals. These residuals, $\check{D}$, represent the part of the treatment that is not linearly predictable from the covariates—the “surprise” in D .

$$\check{D} = D - \text{OLS}(D \sim X).\text{predict}(X)$$

3. **Regress residual on residual:** Run a simple linear regression of the outcome residuals $\check{Y}$ on the treatment residuals $\check{D}$ (without an intercept). The resulting coefficient is $\hat{\theta}$.

$$\hat{\theta} = \text{OLS}(\check{Y} \sim \check{D}) \quad \Leftrightarrow \quad \text{solve } \mathbb{E}_n[(\check{Y} - \hat{\theta}\check{D})\check{D}] = 0$$

Solving the empirical normal equations of the simple line-fit $\text{OLS}(\check{Y} \sim \check{D})$, we can write down the formula for $\hat{\theta}$ explicitly:

$$\hat{\theta} = \frac{\mathbb{E}_n[\check{Y} \cdot \check{D}]}{\mathbb{E}_n[\check{D}^2]} = \frac{\text{Cov}_n(\check{Y}, \check{D})}{\text{Var}_n(\check{D})}$$

Key Insight from FWL

$\hat{\theta}$ is the **normalized correlation** between the residualized outcome and the residualized treatment. It answers the question: “When the treatment is higher than we would expect based on the confounders, is the outcome also higher than we would expect based on the confounders?”

So if you are ever asked: “*how should I interpret the coefficient associated with the first input variable in an OLS specification?*”, the above interpretation is much more intuitive than saying “*it is the first coordinate of $\mathbb{E}[XX']^{-1}\mathbb{E}[XY]$* ”!

This theorem is profound. It tells us that the coefficient on D in a complex regression is simply the line-fit coefficient between the part of Y that X cannot (linearly) explain and the part of D that X cannot (linearly) explain. A complex multivariate regression reduces to a simple line fit on residuals! This perspective is crucial for understanding the conditions under which the OLS

coefficient is uniquely identified and what drives its variance. The FWL theorem will serve as a cornerstone for our exploration of more advanced causal inference techniques, including the double machine learning estimator.

2.2 Python Implementation

The FWL theorem can be verified numerically with a few lines of Python code:

```
# --- Standard OLS approach ---
model_full = LinearRegression().fit(np.c_[D, X], Y)
theta_full = model_full.coef_[0]

# --- FWL three-step procedure ---
# 1. Residualize Y: Regress Y on X
Y_res = Y - LinearRegression().fit(X, Y).predict(X)
# 2. Residualize D: Regress D on X
D_res = D - LinearRegression().fit(X, D).predict(X)
# 3. Final Regression: Regress residualized Y on residualized D
theta_fwl = LinearRegression(fit_intercept=False).fit(
    D_res.reshape(-1,1), Y_res).coef_[0]

# --- Verification ---
print(f"Full OLS theta: {theta_full:.6f}")
print(f"FWL OLS theta: {theta_fwl:.6f}")
# These will be identical!
```

2.3 Visualizing the FWL Theorem

To build intuition, consider a stylized example of estimating price elasticity of demand for a product (ice-cream), where the goal is to measure the effect of price on demand. A simple regression might be confounded by seasonality (Summer vs. Winter); for instance, both price and demand for the product might be high in the summer and low in the winter, leading to a spurious positive correlation (see Figure 1). The FWL procedure corrects for this by first “residualizing” both price and demand against seasonality. This process removes the predictable seasonal variation, leaving only the “surprise” components. More concretely, in this simple case, residualization of the outcome will center all winter demands in the data around the average winter demand and all summer demands around the average summer demand. Similarly, residualization of the treatment will center all winter prices in the data around the average winter price and all summer prices around the average summer price. When these residuals are plotted against each other (see Figure 1), the true negative relationship between price and demand is revealed. This visual demonstration, where the data cloud shifts from a misleading positive slope to the correct negative slope, powerfully illustrates the importance of controlling for confounders and the elegance of the FWL decomposition. The slope of the line in the right panel is the OLS coefficient $\hat{\theta}$ associated with price, when we run an OLS regression of demand on price and seasonality.

The Power of Residualization: From Wrong (0.56) to Right (-1.57)

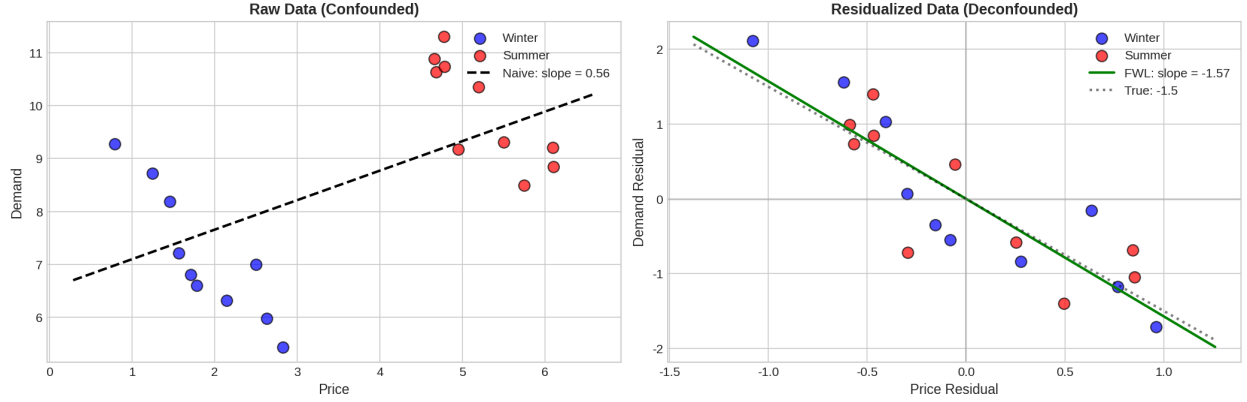


Figure 1: Visualizing the Frisch-Waugh-Lovell theorem. **Left:** In the raw data, the relationship between price and demand is confounded by seasonality. A naive regression (black dashed line) incorrectly shows a positive slope of approximately $+0.56$. **Right:** After residualizing both price and demand with respect to seasonality, the true negative causal relationship emerges. The FWL slope (green line) on the residuals correctly recovers the true effect of -1.57 , very close to the true value of -1.5 (black dotted line). *Data generating process:* 20 samples, binary seasonality confounder, true price effect = -1.5 , price = $2.0 + 3.0 * \text{seasonality} + N(0, 0.5)$, demand = $10.0 + 8.0 * \text{seasonality} + \text{true_effect} * \text{price} + N(0, 0.5)$.

2.4 The FWL Theorem in Population

The FWL theorem also holds in the population limit, where we work with the true Best Linear Predictor (BLP) instead of the empirical OLS approximation.

Theorem (Frisch-Waugh-Lovell in Population)

The coefficient θ on D in the Best Linear Predictor $\text{BLP}(Y \sim D, X)$ is identical to the coefficient from:

1. Calculate BLP of Y using X and compute *population* residuals:

$$\tilde{Y} = Y - \text{BLP}(Y \sim X).\text{predict}(X)$$

2. Calculate BLP of D using X and compute *population* residuals:

$$\tilde{D} = D - \text{BLP}(D \sim X).\text{predict}(X)$$

3. Find coefficient in BLP of $\tilde{Y}$ using $\tilde{D}$:

$$\theta = \text{coeff in BLP}(\tilde{Y} \sim \tilde{D}) \quad \Leftrightarrow \quad \text{solve } \mathbb{E}[(\tilde{Y} - \theta \tilde{D})\tilde{D}] = 0$$

Solving the population normal equations of the final step gives us the formula:

$$\theta = \frac{\mathbb{E}[\tilde{Y}\tilde{D}]}{\mathbb{E}[\tilde{D}^2]} = \frac{\text{Cov}(\tilde{Y}, \tilde{D})}{\text{Var}(\tilde{D})}$$

This is the population quantity that the OLS estimator $\hat{\theta}$ converges to as the sample size grows.

2.5 Answering Questions 1 and 2

Armed with the FWL theorem, we can now provide clear and intuitive answers to our first two questions.

Answer to Question 1: How to interpret θ ?

The coefficient θ is the slope of the line-fit between:

- The part of Y not linearly predictable from X (the “surprise” in outcome)
- The part of D not linearly predictable from X (the “surprise” in treatment)

θ captures how much the (*linearly*) unexplained variation in Y moves with the (*linearly*) unexplained variation in D , after controlling for X .

Answer to Question 2: When is θ uniquely identified?

The formula $\theta = \mathbb{E}[\tilde{Y}\tilde{D}]/\mathbb{E}[\tilde{D}^2]$ has a unique solution if and only if the denominator is non-zero, i.e., $\text{Var}(\tilde{D}) > 0$. This means that the treatment D cannot be perfectly linearly predicted from X —there must be some “surprise” left in treatment. If $D = \gamma'X$ exactly (perfect collinearity), then $\tilde{D} = 0$ and $\text{Var}(\tilde{D}) = 0$. The coefficient θ is undefined (can take many possible values in the optimal solution to the Best Linear Predictor problem)—this is the **collinearity problem**.

The condition for identifiability, i.e., $\text{Var}(\tilde{D}) > 0$, is significantly weaker than the strict overlap requirement in the context of binary treatments, where the probability of receiving treatment must be bounded away from 0 and 1 for all values of the covariates. In the linear regression setting, we only need the treatment to not be a perfect linear function of the covariates on average. This allows for some regions of the data where the treatment might be perfectly predictable, as long as there is sufficient variation in other regions to ensure that the overall variance of the treatment residuals is greater than zero. This insight, derived from the FWL theorem, is crucial for understanding the conditions under which we can obtain a unique and meaningful estimate of a treatment effect in a linear outcome model.

3 Proof of the FWL Theorem

The proof of the Frisch-Waugh-Lovell (FWL) theorem is rooted in the fundamental properties of Ordinary Least Squares (OLS) and the concept of residualization. To understand the proof, it is helpful to consider another interpretation of OLS. The OLS coefficients are chosen to decompose the outcome variable into a linearly predictable part and a residual component. A key property of this decomposition is that the residual is orthogonal to, or uncorrelated with, the input features. This orthogonality condition is equivalent to solving the normal equations or to minimizing the mean squared error.

Key Insight: OLS as Orthogonal Decomposition

Finding the $\text{OLS}(Y \sim D, X)$ coefficients is equivalent to finding a decomposition:

$$Y = \theta D + \beta' X + c + \epsilon$$

where the residual ϵ is empirically orthogonal to $(1, D, X)$, denoted $\epsilon \perp_n (1, D, X)$. This means:

$$\mathbb{E}_n[\epsilon] = 0, \quad \mathbb{E}_n[\epsilon \cdot D] = 0, \quad \mathbb{E}_n[\epsilon \cdot X] = 0$$

Any (coefficients, residual) pair satisfying this orthogonal decomposition is a valid $\text{OLS}(Y \sim D, X)$ solution. Think of it as: OLS extracts everything useful from the predictors. The residual has no remaining correlation with any predictor. Our proof strategy is to show that the FWL procedure produces coefficients satisfying such an orthogonal decomposition.

3.1 The Residualization Operator

With this understanding, the proof of the FWL theorem proceeds by defining a “residualization” operator. This operator takes a variable and subtracts its empirically best linear predictor based on a set of other variables, effectively removing the linearly predictable part.

Definition 1 (Residualization Operator). *For any variable V , define the **residualization operator** as:*

$$\check{V} = V - \gamma'_V X - c_V$$

where γ_V and c_V are chosen so that $\check{V} \perp_n (1, X)$.

A crucial property of this operator is its additivity: the residual (surprise) of a sum of variables is the sum of their residuals (surprises).

Lemma 1 (Additivity of Residualization).

$$\widetilde{A + B} = \check{A} + \check{B}$$

Proof. By the definition of residualization, we can write:

$$\begin{aligned} A &= \gamma'_A X + c_A + \check{A}, & \check{A} &\perp_n (1, X) \\ B &= \gamma'_B X + c_B + \check{B}, & \check{B} &\perp_n (1, X) \end{aligned}$$

Thus we can now also write:

$$A + B = (\gamma_A + \gamma_B)'X + (c_A + c_B) + (\check{A} + \check{B}).$$

Since $\check{A} \perp_n (1, X)$ and $\check{B} \perp_n (1, X)$, we have $\check{A} + \check{B} \perp_n (1, X)$ by linearity of expectation. Therefore, the pair of coefficients $(\gamma_A + \gamma_B, c_A + c_B)$ and residuals $\check{A} + \check{B}$ is a “valid” orthogonal decomposition, and therefore $\widetilde{A + B} = \check{A} + \check{B}$. $\square$

3.2 The Proof

Proof of the FWL Theorem. Start with the OLS($Y \sim D, X$) decomposition:

$$Y = \theta D + \beta'X + c + \epsilon, \quad \text{with } \epsilon \perp_n (1, D, X)$$

Step 1: Residualize both sides. Apply the residualization operator to both sides of the equation:

$$\check{Y} = \theta \check{D} + \beta' \check{X} + \check{c} + \check{\epsilon}$$

Step 2: Simplify. Now observe three key facts:

- $\check{X} = 0$ because X is perfectly linearly predictable from X
- $\check{c} = 0$ because a constant is perfectly linearly predictable from any set of variables
- $\check{\epsilon} = \epsilon$ because $\epsilon \perp_n (1, X)$ means ϵ is already its own residual (it has no component predictable from X)

Step 3: Conclude. Therefore:

$$\check{Y} = \theta \check{D} + \epsilon, \quad \text{with } \epsilon \perp_n \check{D}$$

The orthogonality $\epsilon \perp_n \check{D}$ follows from $\epsilon \perp_n (1, D, X)$ and the fact that $\check{D}$ is a linear combination of $(1, D, X)$. Thus, the pair (θ, ϵ) is a valid OLS($\check{Y} \sim \check{D}$) decomposition, which means θ is the solution to the OLS($\check{Y} \sim \check{D}$) problem! $\square$

4 Variance and Standard Errors—The Insensitivity Theorem

Building on the Frisch-Waugh-Lovell (FWL) theorem, we introduce the concept of the insensitivity theorem, which is crucial for valid statistical inference. The FWL theorem shows that the OLS

coefficient of a variable can be obtained by regressing the residuals of the outcome on the residuals of that variable. The insensitivity theorem takes this a step further by demonstrating that when we calculate the standard error for this coefficient, we can effectively ignore the estimation error from the first-stage regressions used to obtain the residuals. In other words, the final estimate is “insensitive” to the uncertainty in the initial residualization steps.

Theorem (Insensitivity)

When calculating standard errors for $\hat{\theta}$, we can ignore the estimation error in the first two OLS steps. Asymptotically, $\hat{\theta}$ behaves as if we had the true population residuals:

$$\tilde{D} = D - \text{BLP}(D|X), \quad \tilde{Y} = Y - \text{BLP}(Y|X)$$

You might worry: “I estimated those residuals—doesn’t that add uncertainty?” Surprisingly, no!

This is a powerful result, as it simplifies the calculation of standard errors and allows us to treat the residuals as if they were the true, population-level values. This insensitivity is analogous to the property of the doubly robust estimator, where the estimator was insensitive to errors in either the outcome regression or the propensity model (nuisance models). Here, the nuisance quantities are the first stage coefficients in the best linear predictors that are used to define the treatment and outcome residuals.

4.1 Deriving the Asymptotic Distribution

We now turn to our second major result, which concerns the computation of standard errors for $\hat{\theta}$. The insensitivity theorem has a significant practical implication: it greatly simplifies the calculation of standard errors for OLS coefficients. Because the estimation error from the first-stage residualization steps can be ignored, we can proceed as if the residuals were the true, known population residuals.

This means that to calculate the standard error for the coefficient of interest, we only need to consider the final, simple univariate regression of the outcome residuals on the treatment residuals. The standard error can then be calculated using the familiar formula for a simple univariate linear regression. The latter, can be easily characterized with a simple application of the Law of Large

Numbers (LLN) and the Central Limit Thoerem (CLT):

$$\begin{aligned}
\sqrt{n}(\hat{\theta} - \theta) &= \sqrt{n} \left(\frac{\mathbb{E}_n[\tilde{D}\tilde{Y}]}{\mathbb{E}_n[\tilde{D}^2]} - \theta \right) \\
&\approx \sqrt{n} \left(\frac{\mathbb{E}_n[\tilde{D}\tilde{Y}]}{\mathbb{E}_n[\tilde{D}^2]} - \theta \right) \quad (\text{by insensitivity}) \\
&= \sqrt{n} \frac{\mathbb{E}_n[\tilde{D}\tilde{Y}] - \theta \mathbb{E}_n[\tilde{D}^2]}{\mathbb{E}_n[\tilde{D}^2]} \\
&= \sqrt{n} \frac{\mathbb{E}_n[\tilde{D}(\tilde{Y} - \theta \tilde{D})]}{\mathbb{E}_n[\tilde{D}^2]}
\end{aligned}$$

By the Law of Large Numbers, the denominator converges to $\text{Var}(\tilde{D})$. By the Central Limit Theorem applied to the numerator:

Insensitivity Theorem: Asymptotic Distribution

$$\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N \left(0, \frac{\mathbb{E}[\tilde{D}^2(\tilde{Y} - \theta \tilde{D})^2]}{\text{Var}(\tilde{D})^2} \right)$$

The asymptotic variance can be estimated using empirical analogues. Let $\epsilon = \tilde{Y} - \hat{\theta}\tilde{D}$ be the final-stage residual. Then:

$$\hat{\sigma}^2 = \frac{\mathbb{E}_n[\tilde{D}^2(\tilde{Y} - \hat{\theta}\tilde{D})^2]}{\text{Var}_n(\tilde{D})^2} = \frac{\mathbb{E}_n[\epsilon^2 \tilde{D}^2]}{\mathbb{E}_n[\tilde{D}^2]^2}$$

The standard error is:

$$\text{s.e.}(\hat{\theta}) = \frac{\hat{\sigma}}{\sqrt{n}}$$

```

# Final stage residuals
epsilon = Y_residuals - theta_fwl * D_residuals

# Asymptotic variance formula
numerator = np.mean(epsilon**2 * D_residuals**2)
denominator = np.mean(D_residuals**2)**2
sigma_sq = numerator / denominator

# Standard error = sigma / sqrt(n)
se_manual = np.sqrt(sigma_sq / len(Y))

```

Practical Note

This is exactly the **HC0 robust standard error** that OLS packages compute! They don't calculate it this way, but it's numerically equivalent. This formula gives a much more interpretable expression.

4.2 Answering Questions 3 and 4

Answer to Question 3: What drives high variance in $\hat{\theta}$?

From the asymptotic variance of $\hat{\theta}$, i.e. $\sigma^2 = \frac{\mathbb{E}[\tilde{D}^2(\tilde{Y} - \theta\tilde{D})^2]}{\text{Var}(\tilde{D})^2}$, we see two key drivers of high variance:

1. **Small denominator:** If $\text{Var}(\tilde{D})$ is small (treatment is highly predictable from confounders), we divide by a small number $\Rightarrow$ high variance. Little “surprise” in treatment means little to learn from.
2. **Large numerator noise:** If the unexplained part of the outcome ϵ that is not linearly explained by the treatment and controls, $\epsilon = \tilde{Y} - \theta\tilde{D} = Y - \theta D - \beta'X - c$ has high variance, the estimate is noisy.

High linear predictability of treatment from covariates leads to high variance—but this is much weaker than requiring overlap everywhere!

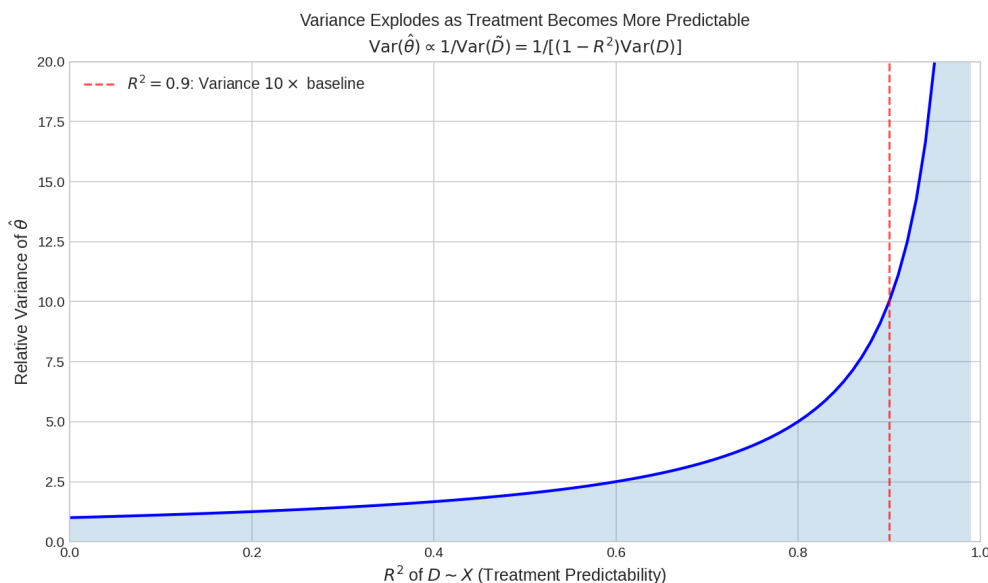


Figure 2: The relationship between treatment predictability and estimator variance. As the R^2 of the regression of the treatment D on the covariates X approaches 1, the variance of the treatment residual $\tilde{D}$ approaches zero, and the variance of the OLS coefficient $\hat{\theta}$ explodes. The formula $\text{Var}(\hat{\theta}) \propto 1/[(1 - R^2)\text{Var}(D)]$ shows this relationship explicitly.

Answer to Question 4: What is the analogue of “overlap”?

The condition is: $\text{Var}(\tilde{D}) > \epsilon$ for some $\epsilon > 0$. We need the residual treatment variance to be bounded away from zero. This is an **average condition**, not a pointwise condition. Even if D is perfectly predictable in some regions of X , we’re fine as long as it’s unpredictable in other regions. Compare to binary treatment overlap ($0 < P(D = 1|X) < 1$ everywhere)—linearity lets us extrapolate!

4.3 Summary: Four Questions Answered

Question	Answer
When is θ uniquely identified?	When D is not perfectly linearly predictable from X (some “surprise” remains)
How to interpret θ ?	Slope of line-fit between residualized Y and residualized D (correlation of surprises)
What drives high variance?	High noise in Y , high linear predictability of D from X (little surprise)
Analogue of overlap?	$\text{Var}(\tilde{D}) > \epsilon > 0$ (average condition, not pointwise)

These answers come from the FWL theorem + Insensitivity theorem!

5 Why Insensitivity Holds

We now provide an intuition for the proof of the insensitivity theorem, which follows the same logic as the insensitivity for the DR estimator. Recall that in the DR setting, we were in a case where our target parameter was equal to some formula $\theta_0 = M(g_0, p_0)$ and then we argued that this formula $M(g_0, p_0)$ was insensitive to the nuisance functions g_0 (outcome regression), and p_0 (propensity), by showing that small perturbations of these functions around their true values, does not change the value of the formula by a lot. We did this by defining an appropriate derivative and showing that this derivative is zero.

5.1 The Insensitivity Argument

We will do exactly the same thing here too. However, in the OLS setting, our target parameter is no longer equal to some formula $M(g_0, p_0)$ but instead it is the solution to an equation $M(\theta; g_0, p_0) = 0$; the normal equation! Our target parameter θ solves the population normal equation:

$$\mathbb{E}[(\tilde{Y} - \theta\tilde{D})\tilde{D}] = 0$$

Note that this equation depends on the nuisance parameters γ_Y and γ_D that correspond to the best linear predictors of Y and D , correspondingly, from X . Define, for arbitrary first-stage coefficients

γ_Y and γ_D the “normal equation”:

$$M(\theta, \gamma_Y, \gamma_D) = \mathbb{E}[(Y - \gamma_Y'X - \theta(D - \gamma_D'X))(D - \gamma_D'X)]$$

and let γ_Y^*, γ_D^* be the true best linear predictors of Y and D , correspondingly. Thus, our target parameter solves the equation:

$$M(\theta; \gamma_Y^*, \gamma_D^*) = 0$$

Claim

The formula $M(\theta; \gamma_Y, \gamma_D)$ is **insensitive** to γ_Y and γ_D around their true values—small errors in these parameters don’t affect substantially the final answer.

We will check this claim, simply by looking at the derivative (gradient) of this formula with respect to these nuisance parameters and showing that it is zero at their true values. It suffices to do this separately for each parameter.

Partial derivative with respect to γ_Y :

$$\partial_{\gamma_Y} M(\theta_0, \gamma_Y^*, \gamma_D^*) = -\mathbb{E}[X(D - \gamma_D^{*\prime}X)] = -\mathbb{E}[X\tilde{D}] = 0$$

The last equality holds because $\tilde{D} \perp X$ by the definition of the Best Linear Predictor (residuals are orthogonal to input features)!

Partial derivative with respect to γ_D : Similarly, one can derive that $\partial_{\gamma_D} M = 0$ by the orthogonality properties that the residuals $\tilde{Y}$ and $\tilde{D}$ need to satisfy, i.e., that $\mathbb{E}[X\tilde{Y}] = 0$ and $\mathbb{E}[X\tilde{D}] = 0$ (prove this as an exercise).

6 Beyond Linearity—Partial Linearity and Double ML

Our exploration of OLS and the Frisch-Waugh-Lovell (FWL) theorem has provided a solid foundation for understanding causal inference in the context of linear models. However, the assumption of full linearity can be quite restrictive in many real-world scenarios. To address this limitation, we can relax our assumptions and move towards a more flexible framework. The partial linear model is a step in this direction. In this model, we maintain the assumption that the outcome is linear in the treatment of interest, but we allow the relationship between the outcome and the confounding variables to be arbitrarily complex and non-linear.

The Partial Linear Model

$$Y = \theta D + f(X) + \epsilon, \quad \mathbb{E}[\epsilon | D, X] = 0$$

In this model, $\mathbb{E}[Y | D, X] = \theta D + f(X)$:

- $f(X)$ can be **any function**—no linearity assumed for how confounders affect outcome

- Only the treatment effect is assumed constant: $\mathbb{E}[Y(d) - Y(0)|X] = \theta d$

This is a significant relaxation, as we are no longer imposing a linear structure on how the confounders affect the outcome. The key assumption is that the treatment effect, represented by θ , is constant across the population and linear. This model is particularly useful when we have a strong reason to believe that the treatment has a simple, linear effect, but we are uncertain about the functional form of the relationship between the confounders and the baseline response. The partially linear model serves as a bridge between the simple linear regression framework and the more complex, fully non-parametric methods that we explored in the binary treatment setting.

6.1 The Generalized FWL Idea

The partially linear model provides a more flexible framework than the fully linear model, and interestingly, it has an exact analogue of the Frisch-Waugh-Lovell (FWL) theorem. A natural question arises: how should the FWL procedure be modified to estimate the treatment effect in this new, more flexible model? The key insight is that instead of using the best *linear* predictor, we should use the best *possible* predictor, which can be a more flexible machine learning model. This means that instead of using OLS to residualize the outcome and the treatment, we can use any suitable machine learning method to predict the outcome from the covariates and the treatment from the covariates.

Theorem (Generalized Frisch-Waugh-Lovell in Population)

The coefficient θ on D in the Partially Linear Model is identical to the coefficient derived from the following procedure:

1. Calculate BP of Y using X , i.e. $\mathbb{E}[Y | X]$ and compute *population* residuals:

$$\tilde{Y} = Y - \mathbb{E}[Y | X]$$

2. Calculate BP of D using X and compute *population* residuals:

$$\tilde{D} = D - \mathbb{E}[D | X]$$

3. Find coefficient in BLP (OLS) of $\tilde{Y}$ using $\tilde{D}$:

$$\theta = \text{coeff in BLP}(\tilde{Y} \sim \tilde{D}) \Leftrightarrow \text{solve } \mathbb{E}[(\tilde{Y} - \theta\tilde{D})\tilde{D}] = 0$$

Proof. The proof of this statement is even simpler than the original FWL theorem. By the definition

of the residuals and the partially linear outcome assumption:

$$\begin{aligned}
\tilde{Y} &= Y - \mathbb{E}[Y|X] && \text{(definition of residual } \tilde{Y}) \\
&= \theta D + f(X) + \epsilon - \theta \mathbb{E}[D|X] - f(X) && \text{(partial linearity)} \\
&= \theta(D - \mathbb{E}[D|X]) + \epsilon \\
&= \theta \tilde{D} + \epsilon && \text{(definition of residual } \tilde{D})
\end{aligned}$$

Since $\mathbb{E}[\epsilon|D, X] = 0$, we have $\mathbb{E}[\epsilon|\tilde{D}] = 0$. Thus, we have:

$$\tilde{Y} = \theta \tilde{D} + \epsilon, \quad \mathbb{E}[\epsilon|\tilde{D}] = 0$$

Which means that θ is the solution to the OLS problem $\tilde{Y} \sim \tilde{D}$. □

Estimation. We can thus replace the Double OLS procedure associated with the original FWL theorem with a Double Machine Learning estimation procedure:

Double Machine Learning Estimation Procedure

The coefficient $\hat{\theta}$ on the treatment variable D in the Partially Linear Model, can be estimated by the following procedure:

1. **Residualize the outcome:** Learn a predictive model $\hat{h}$, predicting the outcome Y from X using **any ML method** and compute the residuals.

$$\check{Y} = Y - \hat{h}(X)$$

2. **Residualize the treatment:** Learn a predictive model $\hat{p}$, predicting the treatment D from X using **any ML method** and compute the residuals.

$$\check{D} = D - \hat{p}(X)$$

3. **Regress residual on residual:** Run a simple linear regression of the outcome residuals $\check{Y}$ on the treatment residuals $\check{D}$ (without an intercept):

$$\hat{\theta} = \text{OLS}(\check{Y} \sim \check{D}) \quad \Leftrightarrow \quad \text{solve } \mathbb{E}_n[(\check{Y} - \hat{\theta}\check{D})\check{D}] = 0$$

This “double machine learning” (DML) approach allows us to account for complex, non-linear relationships between the covariates and both the treatment and the outcome. The residuals obtained from these flexible models represent the “true” surprises in the data, as they capture the variation that cannot be explained by any function of the covariates. By regressing the true surprise in the outcome on the true surprise in the treatment, we can obtain a consistent estimate of the

treatment effect, even in the presence of complex confounding. This generalized FWL procedure forms the basis of the double machine learning estimator, a powerful tool for causal inference in modern data analysis.

6.2 Does Insensitivity Still Hold?

The concept of insensitivity, which we first encountered in the context of the fully linear model, also extends to the more flexible partial linear model. In this setting, the nuisance components are the predictive models for the outcome and the treatment based on the covariates. These can be complex, non-linear machine learning models. The key insight is that the estimating equation for the treatment effect remains insensitive to errors in the estimation of these nuisance functions. This can be shown by taking the functional derivative of the estimating equation with respect to the nuisance functions and demonstrating that it is zero at the true values.

For arbitrary predictive models h and p , define:

$$M(\theta, h, p) = \mathbb{E}[(Y - h(X) - \theta(D - p(X)))(D - p(X))]$$

Let $h_0(X) = \mathbb{E}[Y | X]$ and $p_0(X) = \mathbb{E}[D | X]$. Since $\tilde{Y} = Y - h_0(X)$ and $\tilde{D} = D - p_0(X)$, note that our target parameter is the solution to the equation:

$$M(\theta, h_0, p_0) = 0$$

We can now show that this formula $M(\theta, h, p)$ is insensitive to the nuisance parameters h, p . We will do it separately for each one.

Check sensitivity to h : For any perturbation function $\Delta(X)$, let $h_t = h_0 + t\Delta$

$$\partial_t M(\theta, h_t, p_0)|_{t=0} = -\mathbb{E}[\Delta(X)(D - p_0(X))] = -\mathbb{E}[\Delta(X)\mathbb{E}[D - p_0(X)|X]] = 0$$

Check sensitivity to p : Similarly equals 0.

Yes!

The formula is insensitive to the predictive models—even complex ML models! We can use powerful ML for confounders and still trust our standard errors.

This insensitivity property is a cornerstone of the double machine learning (DML) framework. It allows us to use powerful, flexible machine learning methods to model the complex relationships between the covariates and both the treatment and the outcome, without having to worry about the impact of estimation errors in these models on the final inference for the treatment effect. In the next chapter, this concept will be generalized under the framework of **Neyman orthogonality**. This general theory provides a unified way to understand and derive a wide range of causal estimators that are robust to the use of machine learning for nuisance parameter estimation. The DML

estimator, discussed above, and the Doubly Robust estimator, discussed in prior lectures, are prime examples of this powerful paradigm.

6.3 The Overfitting Problem

Wait!

There's a catch we saw with DR: **overfitting and double-dipping!**

If we use the same data to: 1) Train $\hat{h}$ and $\hat{p}$, 2) Compute residuals, 3) Fit the final regression, then we may overfit and get biased estimates. The model “memorizes” training points, making residuals artificially small.

Solution

Cross-fitting, just like with DR! For each point, predict using a model trained on *other* points.

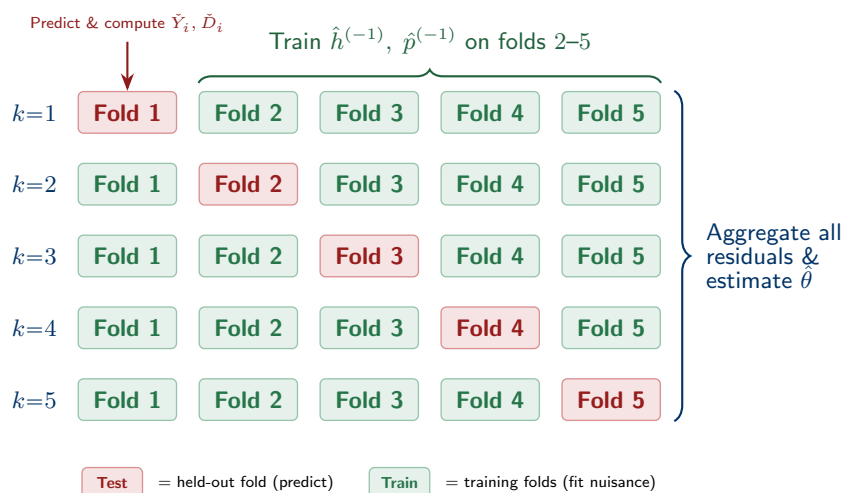


Figure 3: Illustration of 5-fold cross-fitting.

6.4 The Double Machine Learning Algorithm

The double machine learning (DML) algorithm combines the generalized Frisch-Waugh-Lovell (FWL) procedure with the crucial technique of cross-fitting to prevent overfitting. While the generalized FWL approach allows us to use flexible machine learning models to account for complex confounding, a naive implementation can lead to biased estimates due to overfitting. The DML algorithm addresses this by splitting the data into multiple folds. For each fold, the predictive models for the outcome and the treatment are trained on the data from the other folds (out-of-fold) and then used to generate predictions for the data within that fold (in-fold). This ensures that the predictions for each data point are generated by models that were not trained on that data point,

thus avoiding overfitting. Once the out-of-fold predictions are generated for all data points, the residuals for the outcome and the treatment are calculated. The final step of the DML algorithm is to run a simple OLS regression of the outcome residuals on the treatment residuals. The coefficient from this regression is the DML estimate of the treatment effect.

Algorithm (Double ML with K-fold Cross-Fitting)

1. Split data into K folds
2. For each fold k :
 - Train $\hat{h}^{(-k)}(X)$ on out-of-fold data to predict Y from X
 - Train $\hat{p}^{(-k)}(X)$ on out-of-fold data to predict D from X
 - For each sample i in fold k : $\check{Y}_i = Y_i - \hat{h}^{(-k)}(X_i)$, $\check{D}_i = D_i - \hat{p}^{(-k)}(X_i)$
3. Run simple OLS on all residuals (no intercept):

$$\hat{\theta} = \frac{\mathbb{E}_n[\check{Y}\check{D}]}{\mathbb{E}_n[\check{D}^2]}$$

```
from sklearn.model_selection import KFold

def double_ml(Y, D, X, n_folds=5):
    kf = KFold(n_splits=n_folds, shuffle=True)
    Y_res, D_res = np.zeros(len(Y)), np.zeros(len(Y))

    for train_idx, test_idx in kf.split(X):
        # Train on out-of-fold, predict for in-fold
        h_model = AutoML().fit(X[train_idx], Y[train_idx])
        Y_res[test_idx] = Y[test_idx] - h_model.predict(X[test_idx])

        p_model = AutoML().fit(X[train_idx], D[train_idx])
        D_res[test_idx] = D[test_idx] - p_model.predict(X[test_idx])

    # Final line-fit on surprises
    theta_hat = np.mean(Y_res * D_res) / np.mean(D_res**2)
    return theta_hat
```

6.5 Asymptotic Normality of Double ML

This combination of a generalized FWL procedure (which is based on an insensitive formula) and cross-fitting provides a robust and powerful method for causal inference in high-dimensional and non-linear settings. It allows us to leverage the predictive power of modern machine learning methods while still obtaining valid statistical inference for the causal parameter of interest, as the next theorem states.

Theorem

Under regularity conditions, the cross-fitted DML estimator is asymptotically normal:

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N\left(0, \frac{\mathbb{E}[(\tilde{Y} - \theta\tilde{D})^2 \tilde{D}^2]}{\text{Var}(\tilde{D})^2}\right)$$

Required conditions (the ML models need to be “good enough”):

1. **Product rate:** $\sqrt{n} \cdot \text{RMSE}(\hat{h}) \cdot \text{RMSE}(\hat{p}) \rightarrow 0$
2. **Treatment model rate:** $n^{1/4} \cdot \text{RMSE}(\hat{p}) \rightarrow 0$
3. **Consistency:** $\text{RMSE}(\hat{h}) \rightarrow 0$

We saw already in prior lectures how Random Forests, Lasso, and Neural Networks typically satisfy these conditions under small “effective dimension” assumptions.

Based on this theorem, we can estimate the asymptotic variance using the empirical analogue:

$$\hat{\sigma}^2 = \frac{\mathbb{E}_n[(\tilde{Y} - \hat{\theta}\tilde{D})^2 \tilde{D}^2]}{\mathbb{E}_n[\tilde{D}^2]^2}$$

Standard error: $\text{s.e.} = \hat{\sigma} / \sqrt{n}$

```
def double_ml_with_se(Y, D, X, n_folds=5):
    theta_hat, Y_res, D_res = double_ml(Y, D, X, n_folds)

    # Compute standard error
    epsilon = Y_res - theta_hat * D_res
    sigma_sq = np.mean(epsilon**2 * D_res**2) / np.mean(D_res**2)**2
    se = np.sqrt(sigma_sq / len(Y))

    return theta_hat, se
```

6.6 The Overlap Analogue in Partial Linearity

Key Insight

The overlap condition becomes $\text{Var}(\tilde{D}) > \epsilon$. Thus we simply need that the un-predictable part of the treatment $\tilde{D} = D - \mathbb{E}[D | X]$ to have strictly positive variance. This means that the squared residual treatment $\tilde{D}^2$ should just be non-zero **on average**. Much weaker than pointwise overlap!

For binary treatments: this assumption translates to $\text{Var}(\tilde{D}) = \mathbb{E}[p(X)(1 - p(X))]$. We need the propensity to be bounded away from 0 and 1 **on average** and not pointwise for each X . Partial linearity allows us to extrapolate the effect from regions where we have overlap to regions where we don't.

6.7 Software Implementations of the DoubleML Algorithm

EconML (Microsoft)

```
from econml.dml import LinearDML

est = LinearDML(model_y=AutoML(), model_t=AutoML())
est.fit(Y, D, X=None, W=X)
est.summary()
```

DoubleML (Python package)

```
from doubleml import DoubleMLPLR
from sklearn.ensemble import RandomForestRegressor

ml_l = RandomForestRegressor() # outcome regresison
ml_m = RandomForestRegressor()
dml = DoubleMLPLR(data, ml_l, ml_m)
dml.fit()
dml.summary
```

6.8 Industry Applications

Double Machine Learning has been widely adopted in industry for causal inference at scale. For instance, recently Amazon has published research on using DML in their experimentation platform for causal inference at scale.

Paper: “Double Machine Learning at Scale to Predict Causal Impact of Customer Actions” (Amazon Science)

As described in this paper, the Double Machine Learning algorithm is applied to estimate the long-term and short-term effects of user actions on the platform on downstream metrics of interest, after controlling for historical characteristics of the customer. Examples of actions provided in the paper are “what is the effect of purchasing in category X” or “what is the effect of program Z”. 100s of customer actions are being evaluated using the Double ML approach for their effect on downstream metrics.

The DoubleML algorithm has been used in several key industry applications, such as:

- Price elasticity estimation
- Marketing attribution

Why DML? It handles high-dimensional covariates, allows flexible ML models, provides valid inference, works with both continuous and binary treatments (and even many simultaneous treatments), has lower variance than DR, and is less sensitive to overlap violations!

7 Part VI: Interpretation with Treatment Effect Heterogeneity

What does DML estimate if the conditional expectation function (CEF) is not partially linear?

7.1 Treatment Effect Heterogeneity: Binary Case

Even though, we motivated the DoubleML algorithm using continuous treatments, the algorithm can be applied as stated even in the case when the treatment D is binary. In this case, the partial linearity assumption simply becomes the assumption that the treatment effect is constant and does not depend on X , i.e. $\mathbb{E}[Y(1) - Y(0) | X] = \theta$. Under this assumption the DoubleML algorithm estimates the ATE and is an alternative to the DR algorithm we have already seen in prior lectures.

What if the true model has heterogeneous effects? What does the DoubleML procedure estimate? Suppose, (without loss of generality for binary treatments) that

$$\mathbb{E}[Y|D, X] = \tau(X) \cdot D + f(X)$$

where $\tau(X)$ varies with X and corresponds to the heterogeneous treatment effect

$$\tau(x) = \mathbb{E}[Y(1) - Y(0) | X = x].$$

We can still run the DML procedure. Irrespective of partial linearity, DML will always converge to the quantity:

$$\theta_{\text{DML}} = \frac{\mathbb{E}[\tilde{D}\tilde{Y}]}{\text{Var}(\tilde{D})}, \quad \text{with } \tilde{Y} = Y - \mathbb{E}[Y|X] \text{ and } \tilde{D} = D - \mathbb{E}[D|X]$$

After simple numerical manipulations we can argue that this quantity corresponds to an interpretable weighted average effect (prove this as an exercise):

Overlap-Weighted Average Treatment Effect

DML estimates a weighted average of $\tau(X)$, with weights $\propto \text{Var}(D|X)$ —more weight where treatment is more “random” (less predictable):

$$\theta_{\text{DML}} = \frac{\mathbb{E}[\tilde{D}\tilde{Y}]}{\text{Var}(\tilde{D})} = \frac{\mathbb{E}[\tau(X)\text{Var}(D|X)]}{\mathbb{E}[\text{Var}(D|X)]}$$

Why Use DML Even for Binary Treatments? Even though DR works for binary treatments, DML has advantages:

1. **Lower variance:** Weights by $\text{Var}(D|X) = p(X)(1 - p(X))$, downweighting extreme propensities
2. **Natural interpretation:** Overlap-weighted ATE
3. **Robustness to overlap violations:** Works even with some regions of poor overlap

Trade-off

Has lower variance and does not require overlap, but requires partial linearity assumption (constant treatment effect). If effects are heterogeneous, DML gives a weighted average, not the population ATE that DR gives.

7.2 Continuous Treatments: Weighted Average Derivatives

For continuous treatments **without** partial linearity, let $g(d, X) = \mathbb{E}[Y|D = d, X]$. How can we interpret the DoubleML parameter θ_{DML} even in the absence of the partial linearity assumption? Even without assuming $g(d, X) = \theta d + f(X)$, DML estimates a **weighted average marginal effect**:

$$\theta_{\text{DML}} = \frac{\mathbb{E} \left[w(D, X) \cdot \frac{\partial}{\partial d} g(d, X) \Big|_{d=D} \right]}{\mathbb{E}[w(D, X)]}$$

for a non-negative weight function $w(D, X) > 0$. Think of it as: “A weighted average of how much the outcome changes when we nudge treatment?” The weights are complex and unintuitive in general, i.e.,

$$w(d, x) = \frac{\mathbb{E}[\tilde{D} 1\{D \geq d\} \mid X = x]}{f(d \mid x)}$$

where $f(d \mid x)$ is the conditional density of the treatment given the controls. Even so, DML still has a causal interpretation as a non-negative weighted average derivative!

Special Case: Gaussian Treatment When $D|X \sim N(p(X), \sigma^2(X))$, the story simplifies beautifully. The DML numerator is:

$$\mathbb{E}[\tilde{Y} \cdot \tilde{D}] = \mathbb{E}[Y \cdot \tilde{D}] = \mathbb{E}[g(D, X) \cdot (D - p(X))]$$

By **Stein’s Lemma** (a result for Gaussian random variables):

$$\mathbb{E}[g(D, X) \cdot (D - p(X))] = \mathbb{E} \left[\sigma^2(X) \cdot \frac{\partial}{\partial d} g(d, X) \Big|_{d=D} \right]$$

Thus, we can interpret the DML coefficient as a more interpretable variance-weighted average derivative with weights $w(d, x) = \sigma^2(x)$:

Variance-Weighted Average Marginal Effect

$$\theta_{\text{DML}} = \frac{\mathbb{E}[\sigma^2(X) \cdot \partial_d g(D, X)]}{\mathbb{E}[\sigma^2(X)]}$$

where:

- $\partial_d g(D, X)$ is the marginal effect at (D, X)
- $\sigma^2(X)$ is the conditional variance of treatment given X

Hence, we weight the marginal effect by the variance: more weight where treatment is more “randomized”. This is almost the exact analogue of the overlap-weighted average treatment effect from the binary case. This intuition should be approximately correct even when the treatment is approximately Gaussian conditional on the controls. Even without partial linearity, DML has a clean causal interpretation!

8 Summary: Four Key Takeaways

This chapter provided a deep dive into the mechanics of OLS and used it as a springboard to develop the powerful Double Machine Learning estimator. Our four goals correspond to four key takeaways:

1. **The Frisch-Waugh-Lovell Theorem:** The coefficient θ in $\text{OLS}(Y \sim D, X)$ equals the slope from regressing the “surprise” in Y on the “surprise” in D —the normalized correlation of residuals.
2. **Insensitivity to First-Stage Errors:** When computing standard errors for θ , we can ignore estimation error in the first-stage regressions—another “insensitive formula” like the DR estimator.
3. **Partial Linearity Enables Flexibility:** Assuming $\mathbb{E}[Y|D, X] = \theta D + f(X)$ allows ML for $f(X)$ while still getting valid inference on θ .
4. **Double ML Estimator:** DML estimates average treatment effects (or average marginal effects) under partial linearity—widely used in industry.

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